MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6832

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Twin triode, subminiature, medium Mu, dc amplifier

Outline --- 3-2 (EIA)

Base --- E8-10

Envelope --- T3

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	2a	2g	h	2k	1k	h	1g	1a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ehk	Rk/k	Rg/g	Ib/a	Ic/g	Pp/a	TE	Alt
Unit:	V	Vdc	Vdc	v	Ohms	Meg	mAdc	mAdc	W	°C	ft
Maximum:	6.6	165	0, -20	100	---	10	3.0	0.1	0.1	140	Note 1
Minimum:	6.0	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	100	0	---	3,000	---	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Heater current		0.4	II	If	370	430	mA
1336	Heater-cathode leakage	See note 2	0.4	II	Ihk	---	10	µAdc
1266	Total grid current	See notes 2 and 3	0.4	II	Ic	0	-0.03	µAdc
1256	Electrode current (1) (anode)	See note 2	0.4	II	Ib	0.6	1.0	mAdc
1256	Electrode current (2) (anode)	Ec1 = -4.5 Vdc (see note 2)	0.4	II	Ib	---	100	µAdc
1306	Transconductance (1)	See note 2	0.4	II	Sm	850	1,250	µmhos
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
(D) 1211	Insulation of electrodes		---	---	---	---	---	---
---	Mechanical stability	See figure 1 and notes 4 and 5	2.5	II	ΔEp-a	---	0.5	Vdc
---	Electrical stability	See figure 1 and notes 4 and 6	2.5	II	ΔEp-a	---	0.3	Vdc
1031	Low-frequency vibration	Rp = 10,000 ohms; F = 40 Hz; 15 G (see note 2)	---	---	Ep	---	10	mVac
1316	Amplification factor	See note 2	---	---	Mu	22	30	---
1256	Electrode current (1) (anode) (difference between sections)		6.5	I	ΔIb	---	0.15	mAdc
1306	Transconductance (2)	Ef = 5.7 V (see note 2)	---	---	ΔSm Ef	---	10	%
(D) 1246	Audio frequency noise	Eca1 = 45 mVac; Rg = 0.5 Meg; Rp = 0.2 Meg; Rk = 1,500 ohms (see note 7)	2.5	I	EB	---	17	vu
---	Drift (see figure 1)	t = 30 sec (see notes 4 and 8)	---	---	(1)ΔEp-a	---	1.2	Vdc
---	Drift (see figure 1)	t = 15 min (see notes 4 and 8)	---	---	(2)ΔEp-a	---	0.6	Vdc
1116	Lead fatigue		2.5	Code F	---	---	---	---
1041	Shock (1)	450 G (see note 9)	---	---	---	---	---	---
1031	Vibration-fatigue test	2.5 G; fixed frequency F = 25 min; 60 max	6.5	See note 10	---	---	---	---

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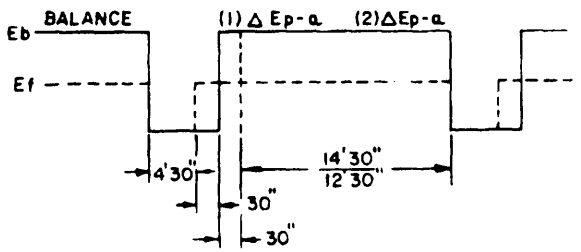
METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
---	Post shock (1) and vibration-fatigue test	Low-frequency vibration	---	---	Ep	---	40	mVac
		Heater-cathode leakage	---	---	Ihk	---	40	μAdc
		Transconductance (1)	---	---	Sm	750	---	μmhos
		Total grid current	---	---	Ic	0	-0.1	μAdc
---	Shock test (2)	75 G (hammer angle = 120° + rubber pad) t = 10 ms duration (see notes 11 and 12)	---	---	---	---	---	---
---	Post-shock test (2) end points	Change of anode voltage (ΔEp-a) of the two units (see figure 1 and note 12)	---	---	ΔEp-a	---	1.2	Vdc
2126	Glass strain		4.0	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC 1 ST SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC COMBINED SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3									
1506	Heater-cycling life test	Ef = 7.0 V; 1 min "on", 4 min "off"; Ehk = 140 Vac; Ec1 = Eb = 0	---	---	---	---	---	---	---	---
1521	Survival-rate life test	TA = room; Rg = 1.0 Meg	---	---	---	---	---	---	---	---
---	Survival-rate life-test end points	Short and discontinuity detection	---	---	---	---	---	---	---	---
		Transconductance (1)	---	---	---	---	ΔSm_t	---	10	%
1501	Intermittent life test	$\overline{TE}$ = 140°C (min); Rg = 1.0 Meg (see note 13)	---	---	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours)	Inoperatives	---	---	1	3	---	---	---	---
		Total grid current	---	---	1	3	Ic	0	-0.05	μAdc
		Transconductance (1)	---	---	1	3	ΔSm_t	---	15	%
		Transconductance (2)	---	---	2	5	ΔSm_{Ef}	---	10	%
		Total defectives	---	---	4	8	---	---	---	---

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. Test each unit separately.
3. This test to be performed at the conclusion of the holding period.
4. At the start of the test and after a minimum warmup period of 5 minutes with Ef = 6.3 V, an initial adjustment of grid bias shall be made to balance the anode voltages of the individual sections within 0.1 Vdc. This voltage unbalance is defined as Ep-a.
5. Tube shall be subjected to a minimum of six repeated shocks of approximately 80 G and approximately 0.2-millisecond duration. The six shocks shall be comprised of two shocks in each of three horizontal directions spaced approximately 120°.
6. Allow tube to warmup for a minimum of 5 minutes in test circuit with 6.3 volts on the heater, then balance the output and change the heater voltage to 5.9 volts. After 2 minutes, read the voltage unbalance.
7. Tie 1k to 2k; 1g to 2g; and 1a to 2a.
8. Operate tubes for 6 cycles. The first cycle shall start after initial adjustment is made and each cycle shall consist of a 5-minute "off" and a 15-minute "on" period of anode voltage applied. Heater voltage shall be applied 30 seconds before the start of each 15-minute "on" period. Readings of voltage unbalance shall be taken during the first and sixth "on" periods at 30 seconds, and at 13 to 15 minutes. The tube shall not be rebalanced at any time after the initial adjustment.



9. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is used.
10. (D) This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
11. The provisions of method 1041 shall apply, except for test conditions listed for shock test (2).

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NOTES: -Continued

12. The measurement of change of anode voltage unbalance between sections (ΔEP-a) of the two units shall be performed as follows:

    (a) The tube shall be preheated for a minimum of 5 minutes, with Ef = 6.3 V, in the circuit on figure 1.

    (b) The 10-turn precision potentiometer (Pot. 1) shall be adjusted to provide an anode voltage unbalance Ep-a in the two units not greater than 0.1 Vdc.

    (c) Record the potentiometer dial setting of b above.

    (d) The tube is then removed from the test circuit and subject to shock test (2).

    (e) After the procedure in (d) above, the tube shall again be preheated for a minimum of 5 minutes with Ef = 6.3 V in the circuit on figure 1, and with the potentiometer at the setting recorded in (c) above.

    (f) The anode voltage unbalance between sections, ΔEp/a, shall be within the limits specified.

13. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 17
DSA - ES

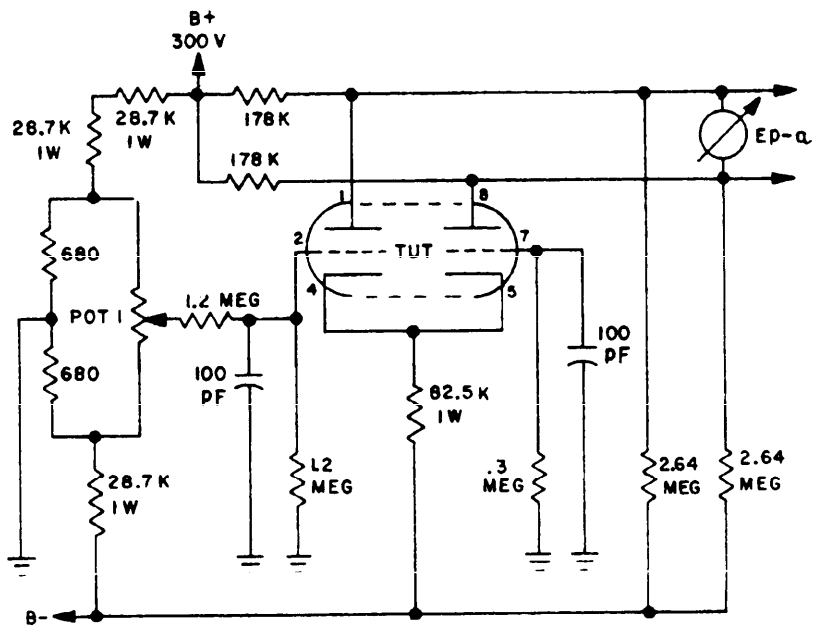
User activities:
Army - WC
Navy - AS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

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NOTE:
Pot. 1 consists of a 10-turn 5,000-ohm precision potentiometer in parallel with a 5,000-ohm .25 percent one-half watt resistor.

FIGURE 1. Test circuit.

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NAVAL ELECTRONIC SYSTEMS COMMAND
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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